

UTSAV GANDHI

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Summary

Hands-on AI/ML practitioner with a Python-first background across classical machine learning, generative AI, and applied data science. Built and evaluated ML models (Random Forest, SVM, KNN, Logistic Regression) during a machine learning internship, and developed GenAI systems including a retrieval-augmented document Q&A pipeline (LangChain, Mistral AI, ChromaDB) and multi-pattern LangGraph agent workflows. Comfortable with the standard ML/data stack (NumPy, Pandas, scikit-learn, TensorFlow) and data structures and algorithms, with 78+ problems solved on LeetCode in Python. Pursuing an M.E. in Cyber Security at Gujarat Technological University, with dissertation research applying AI-safety principles to multi-agent LLM pipelines.

Technical Skills

Data Structures & Algorithms: Arrays, Strings, Hash Tables, Sorting, Math, Dynamic Programming, Backtracking - 78+ • Data structures and algorithms(DSA) problems completed on Leetcode (Python), NeetCode submissions on GitHub

Machine Learning: scikit-learn, TensorFlow, Pandas, NumPy, Logistic Regression, Random Forest, SVM, KNN, Decision Trees, Model Evaluation (F1/Precision/Recall/MSE/MAE)

NLP & GenAI: LangChain, LangGraph, CrewAI, Sentence-Transformers (BERT/SBERT), Prompt Engineering, Retrieval-Augmented Generation, Groq API, Hugging Face, Mistral AI, Model Context Protocol (MCP)

Vector Search & Retrieval: FAISS, ChromaDB, Semantic Search, MMR Retrieval

AI Safety & Evaluation: Prompt Injection Defense, Guardrail Design, pytest-based Eval Harnesses, Adversarial Benchmarking (InjecAgent, ASB, Agent-SafetyBench)

Deployment & Serving: FastAPI, Streamlit, Docker, Vercel, REST API Design

Languages & Web: Python, JavaScript, TypeScript, React.js, Next.js, Django, Node.js

Databases: PostgreSQL, MongoDB, SQLite, Redis, SQL

Security Fundamentals: OWASP Top 10, VAPT Fundamentals, SIEM (Wazuh)

Cloud Certifications: AWS (Generative AI, Prompt Engineering with Claude)

Experience

Full Stack Developer

Feb 2025 – May 2025

The Special Character, Ahmedabad, Gujarat, India

- Engineered both front-end (Next.js 15, React 19) and back-end (PostgreSQL, MongoDB, NeonDB with Redis caching) components of a production web application, cutting API latency by 40% and reducing primary database load by 60%.
- Accelerated page-load speed by 30% across devices through responsive UI improvements.
- Delivered and validated RESTful APIs with Postman, integrating Auth.js v5 authentication in an Agile/Scrum workflow.
- Partnered with a cross-functional engineering team to integrate frontend and backend architectures and verify RESTful API dependencies.

Machine Learning Intern

Jul 2024 – Aug 2024

Innvonix Tech Solutions Pvt. Ltd., Ahmedabad, Gujarat, India

- Trained and evaluated two ML models - credit card fraud detection and home price prediction - comparing Random Forest, KNN, Decision Tree, and SVM classifiers, tuning hyperparameters to reach 86–90% accuracy on test data.
- Benchmarked model performance using F1-score (0.82), precision, and recall for classification and MSE/MAE for regression, rather than accuracy alone.
- Reduced data-preprocessing time by 80% by engineering feature-extraction and EDA pipelines in Python/Pandas.
- Applied advanced Python OOP principles - inheritance, encapsulation, polymorphism, abstraction, decorators, and generators - to build 20+ hands-on programming exercises.

Projects

CyberTrace AI - Financial Fraud & Intelligence Platform

Jul 2026 – Present

3rd Runner-Up, KANAD S.H.I.E.L.D. 2026 Grand Finale, Ahmedabad Cyber Crime Branch

- Developed complex, real-time React.js front-end interfaces, including a live-updating threat-intelligence dashboard with a 2-second refresh interval, using Framer Motion for smooth animated state transitions.
- Visualized deterministic graph analytics across 3 techniques for mule-account detection - DFS-based cycle detection, layering/smurfing path tracing, and fan-in/fan-out account classification - using React Force Graph 2D.
- Extracted entities from FIRs and bank statements across English, Hindi, and Gujarati using a multi-tier OCR pipeline (PyMuPDF, Tesseract, YOLOv8) achieving 90–96% accuracy, including UPI IDs and IFSC codes.
- Secured chain-of-custody integrity with SHA-256 hash generation on every uploaded document, alongside 3 additional safeguards - CSP, stateless JWT authentication, and input sanitization.

Tech Stack - Backend: Django, Django REST Framework, SQLite3, ChromaDB, Django Q2, LangChain, LangGraph, DeepAgents, YOLOv8, Tesseract OCR

Tech Stack - Frontend: React 18, TypeScript, Vite, Zustand, Framer Motion, React Force Graph 2D, Recharts, Tailwind CSS

RAG-Based AI Book Assistant Jul 2026 – Aug 2026

- Built a PDF ingestion pipeline with overlapping-context chunking and Mistral-generated embeddings stored in a persistent ChromaDB vector store.
- Grounded generated answers strictly in source documents using MMR (Maximal Marginal Relevance) retrieval, reducing hallucinated responses and reaching 98% response accuracy.
- Shipped a Streamlit front-end enabling PDF upload and natural-language Q&A over the ingested document.

Tech Stack: LangChain, Mistral AI, ChromaDB, PyMuPDF, Streamlit

Agentic AI with LangGraph AY 2025–2026

- Orchestrated 5 distinct LangGraph workflow patterns - sequential pipelines, parallel fan-out/fan-in with a custom state reducer, conditional intent-based routing, and self-reflection loops with external tool calling via Tavily search.
- Automated a human-in-the-loop approval flow using LangGraph interrupts and MemorySaver state persistence, pausing execution for review before resuming.
- Enabled a Streamlit-based college-assistant application that routes student queries across 3 domain-specific FAISS retrievers - Academic, Rules, and Fee - through an intent-classification node, backed by Groq and Mistral LLMs.

Tech Stack: LangGraph, LangChain, Groq, Mistral AI, HuggingFace Sentence Transformers, FAISS, Tavily, Streamlit

Education

Master of Engineering (M.E.) - Cyber Security Jul 2025 – Jul 2027
Gujarat Technological University (GTU-SET), Ahmedabad, India CGPA: 9.30 / 10

Dissertation: CIRRUS - A Cross-Layer Defense Framework for Multi-Agent LLM Pipelines

Bachelor of Engineering (B.E.) - Information Technology (Specialization in Data Science) Aug 2021 – Jun 2025
SAL Engineering and Technical Institute, Ahmedabad, India CGPA: 8.96 / 10

Certifications

- **AWS** Sep 2026
Fundamentals of Generative AI; Fundamentals of Machine Learning and Artificial Intelligence; Fundamentals of Prompt Engineering with Claude; Exploring Artificial Intelligence Use Cases and Applications - four foundational badges spanning generative AI, core ML/AI concepts, prompt engineering for Claude models, and applied AI use cases.
- **AI Security University - Introduction to AI Security** Sep 2026
Covered AI system attack surfaces, adversarial threats, and defensive design patterns for securing ML/LLM applications.
- **TryHackMe - Advent of Cyber 2025** Dec 2025
Completed daily hands-on security challenges covering web exploitation, log analysis, and incident-response fundamentals.
- **HackerRank - SQL (Advanced)** Aug 2024
Validated advanced SQL proficiency, including complex joins, subqueries, window functions, and query optimization.
- **HackerRank - Python (Basic)** Aug 2024
Solved Python problems spanning basic to intermediate difficulty with clean, optimized code.
- **IBM - Artificial Intelligence Fundamentals** Jul 2024
Core AI/ML concepts, common machine-learning workflows, and practical use cases.
- **Cisco - Data Analytics Essentials** Jul 2024
Data-analysis fundamentals: data collection, cleaning, visualization, and interpretation.
- **Google - Foundations of Cybersecurity** Mar 2024
Core security concepts, common threat types, security frameworks, and the role of a security analyst.